

Elio YE

📍 London, UK 📞 +44 7594 569253 ✉ yyp6714@gmail.com 🌐 elio721.github.io/Elio

EDUCATION

University College London (UCL)

London, UK

BSc Mathematics

09/2024 – Expected 06/2027

Second-year average score: 73. First-year grade: 2:1.

Relevant modules: Introduction to Programming; Complex Analysis; Further Linear Algebra; Fluid Mechanics; Mathematical Methods; Probability and Statistics; Computational Methods and Programming.

TECHNICAL SKILLS

Programming: Python (NumPy, pandas, Matplotlib), Lean
Machine Learning: PyTorch, scikit-learn, CNNs, CSP-LDA, cross-validation, regularisation
Signal Processing: MNE, EEG preprocessing, band-pass filtering, epoching, time-series classification
Mathematics / Tools: Linear Algebra, Probability and Statistics, Analysis; Git/GitHub, Kaggle, LaTeX

PROJECTS

Deep Learning Approaches for Non-invasive EEG-based Brain-Computer Interface
Independent Project 06/2026 – 08/2026

- Developed an end-to-end **four-class motor-imagery EEG classification pipeline** using the BCI Competition IV 2a dataset, processing recordings from **9 subjects** with MNE and Python.
- Applied EEG preprocessing and **8–30 Hz band-pass filtering**, and implemented a **CSP-LDA** pipeline with cross-validation for spatial feature extraction and classification, achieving **67.1% mean subject-level accuracy**.
- Implemented an **EEGNet-based convolutional neural network** in PyTorch and systematically compared deep-learning and classical approaches across subjects.
- Analysed **inter-subject variability**, confusion matrices and training/validation behaviour, identifying overfitting in EEGNet and investigating dropout, weight decay and early stopping to improve generalisation.

UCL Summer Project: Lean Proof Assistant

05/2025 – 07/2025

- Explored formal mathematical proof and verification with the Lean proof assistant.
- Formalised analysis theorems, including a proof that every bounded monotonic sequence converges, and gained experience in computer-assisted reasoning.

Kaggle Competition – Digit Recognizer

08/2025 – 09/2025

- Achieved a score of **0.99346** on the MNIST Digit Recognizer competition.
- Built a CNN with data augmentation, batch normalisation and dropout; applied 5-fold cross-validation, model ensembling, early stopping and learning-rate scheduling.

ADDITIONAL INFORMATION

Languages: English (fluent); Mandarin and Cantonese (native).

Work eligibility: Eligible to work in the UK out of term time.